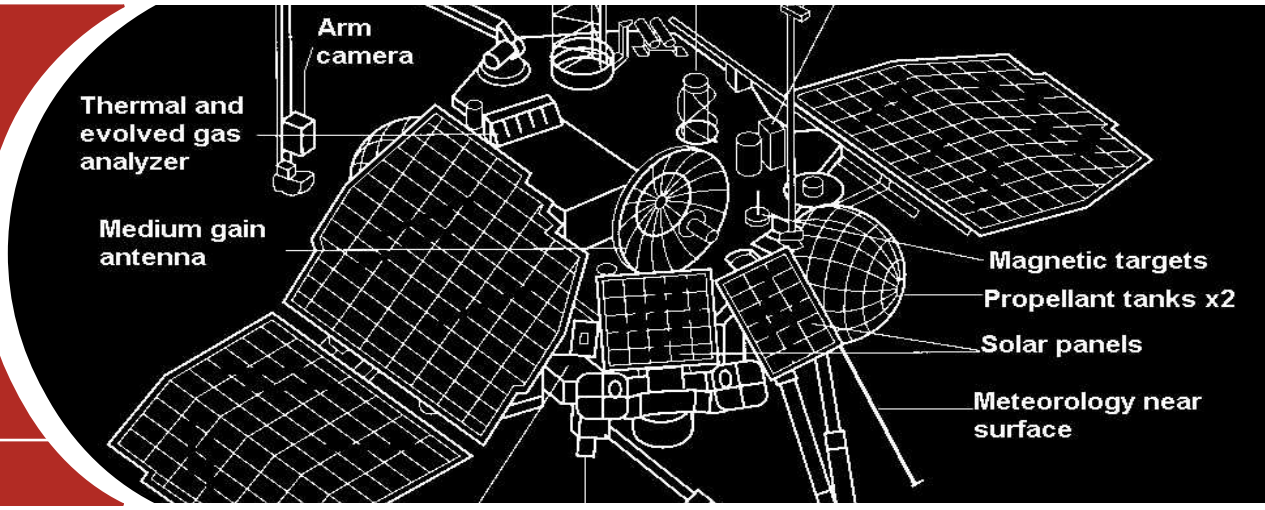
National Aeronautics and Space Administration



NASA SAFETY CENTER
SYSTEM FAILURE CASE STUDY

September 2012 Volume 6 Issue 8

Lost Lander

Failure of the Mars Polar Lander

PROXIMATE CAUSE

- Spurious signals may have been generated when the lander legs were deployed during descent, resulting in premature shutdown of the lander engines

UNDERLYING ISSUES

- Design, Review, and Verification and Validation
- Project Implementation and Demand Margins

AFTERMATH

- Creation of a dedicated JPL Mars Program Office that reported directly to the Laboratory Director
- Creation of an independent organization at the Directorate level dedicated to implementing major flight projects
- Recognition of the FBC concept as relevant and recommendation as the continued approach utilized in the Mars Program, with the stipulation that FBC should not compromise safety

January 3, 1999: The Mars Polar (MPL) was launched from Cape Canaveral, FL, and began an 11-month cruise to Mars, where it was to study soil and climate history in the southern polar Planum Australe region during its 90-day mission. MPL approached Mars on December 3, 1999, according to schedule and after final trajectory corrections, achieved correct entry attitude for descent into the Martian atmosphere. An accepted risk by the project, the MPL antennae would not be aligned with Earth, and an expected loss of communication would occur during the entry, descent, and landing (EDL) sequence. Controllers expected to reestablish communication at 20:39:00 Coordinated Universal Time (UTC), after MPL had landed. However, all attempts at reestablishing communication with the lander failed.

Background

Design

Encased in an aerodynamic entry vehicle comprising a forward heatshield and a backshell, the lander separated from the cruise stage of its launch vehicle approximately 5 minutes before atmospheric entry. During the EDL sequence, MPL was designed to deploy a parachute and jettison its heatshield. During descent, MPL would then release two Deep Space 2 (DS2) microprobes as part of an add-on mission designed to collect soil, meteorology, and seismic data. After release of the DS2 microprobes, MPL would deploy its landing legs and confirm ground acquisition by radar.

After acquisition, the lander was designed to separate from its backshell and parachute and begin powered descent to the surface. The complete EDL sequence took approximately 5.5 minutes to complete. After landing, the solar-powered MPL was to send data to the Mars Climate Orbiter (MCO) for relay to Earth; however, the failure of MCO on September 23, 1999, left MPL to use its x-band radio and medium-gain antenna (MGA) in conjunction with the Deep Space Network (DSN) antenna array to communicate with operators on Earth.

What Happened

Without communication and EDL geometry data, it was impossible to know whether

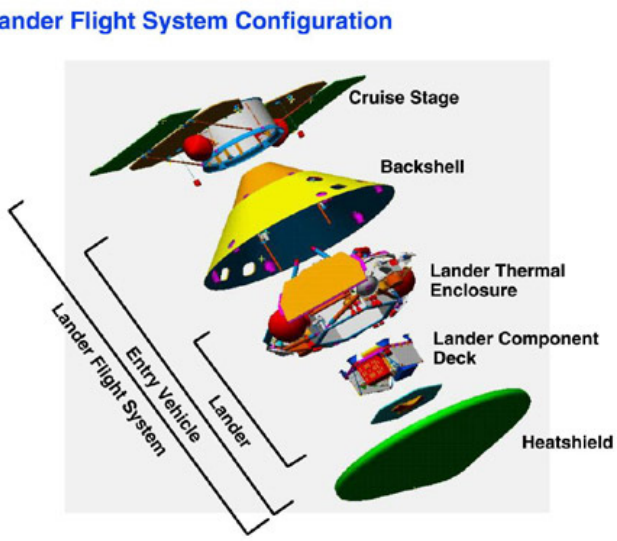


Figure 1. Arrangement of MPL landing components. Source: NASA

MPL reached its powered descent propulsion phase. If it did reach this phase, one possible failure scenario illustrated premature engine shutdown resulting in a crash landing.

Also of note, the DS2 microprobes—designed to impact with the Martian surface without using a device to slow their rate of descent—were lost after communication attempts with the microprobes failed.

Proximate Cause

Independent investigators determined that a key issue that may have caused the failure was spurious signals generated when the lander legs were deployed during descent. The spurious signals gave a false indication of a landing, resulting in premature shutdown of the lander engines and the destruction of the lander when it crashed into the Martian surface. This finding was not confirmed because of the lack of data transmitted during EDL.

Underlying Issues

While software and hardware design played a major role in the failure of MPL, they correlate with project implementation concerns and program-level communication issues.

Design, Review, and Verification and Validation

At the time MPL was designed and built, there was a known problem for sensors involved with mechanical operations, such as lander leg deployment, to produce spurious signals. For MPL, no software requirement existed to clear spurious signals prior to the lander using the sensor information to confirm successful landing. During tests of the lander system, the sensors were incorrectly wired because of a design error. As a result, the systems test did not identify spurious signals and was not repeated with properly wired touchdown sensors. Furthermore, the Jet Propulsion Laboratory (JPL), which managed the MPL project, noted in its investigation report that certain failure modes (e.g., transient signals) were not specifically outlined in the system-level requirement document. While the most probable cause of the failure was premature engine shutdown, an underlying issue was inadequate software design and testing.

Project Implementation and Demand Margins

From the onset, the MPL project, a component of the overarching Mars '98 project, was under significant cost risk, partly because of an aggressive proposal from contractor Lockheed Martin Astronautics (LMA). Adding to this situation, JPL believed that no cost increase was possible, even after MPL scientific payload requirements began to creep. Investigators later estimated at least a 30-percent underfunding.

In contrast, the highly successful and well-managed Pathfinder mission possessed a project management budget of $11 million while the entire Mars '98 program (including MCO [which also failed] and MPL) allocated only $5 million. The Pathfinder budget for Science and Instrument Development (SID) was $14 million, compared with the overall $37 million SID budget for Mars '98.

In 1999, JPL was tasked with a considerable number of projects; the Mars Program Independent Assessment Team (MPIAT) described a tremendous workload on JPL management. As a result of a 3-year overhaul to the Faster, Better, Cheaper (FBC) initiative in the late 1990s, JPL had gone from averaging 1 to 4 projects in a given year to a level of 10

REVIEW AND ANALYZE RECENT MARS AND DEEP SPACE MISSIONS

Pathfinder–Mars '98 Development Cost Comparison

(1999 $ M)

Element	Pathfinder	Mars '98 (MCO & MPL)
Project Management	11	5
Mission Engineering and Operations Development	10	6
Flight System	134	133
Science and Instrument Development	14	37
Rover	25	
Other	2	7
Total	196	188

Figure 2. Although the Pathfinder mission involed an active rover, the Mars '98 project suite's SID budget was more than twice that of Pathfinder, utilizing less than half of Pathfinders Project Mangement budget. Source: NASA.

to 15 concurrent projects. This profusion of projects translated into an understaffed project office. The MPL project brought in highly competent, yet inexperienced project managers, a personnel scenario that called for significant involvement by senior management to compensate for the lack of experience. However, given so many projects, senior management could not focus its attention completely on MPL, and the result was an inadequate application of JPL institutional capability through the period of transition to FBC. Insufficient checks and balances in MPL development and testing resulted from this inattention.

The MPL staff faced tight margins including a limited selection of launch vehicles, growth in the MPL scientific payload requirements, and a fixed planetary launch window. The sum of these resulted in analysis and testing deficiencies and inadequate preparation for mission operations.

The MPIAT found that "JPL management was attempting to convey general program advocacy and to promote a positive customer relationship, motivated by fear of losing business. The result was that JPL management did not convey an adequate risk assessment to NASA Headquarters. NASA Headquarters interpretation was that JPL was agreeing with and accepting objectives, requirements, and constraints. This communication dynamic prevented open and effective discussion of problems and issues. JPL management did not effectively express their concerns to NASA Headquarters about programmatic constraints, and NASA Headquarters did not seem receptive to receiving bad news. Consequently, frank discussions identifying and managing program risks and problems did not occur."

The communication dynamic bypassed the discussion of further testing due to project funding concerns; testing may have identified the software issue regarding identifying and clearing spurious signals.

Figure 3. The Boeing Integrated Defense Systems Delta II 7425 launch vehicle was chosen to send to MPL to Mars; MPL is pictured being placed in the payload fairing of the vehicle. Source: NASA

Figure 4. KSC technicians check underneath MPL as it sits on a workstand in a laboratory cleanroom. Source: NASA

Aftermath

Formation of the MPIAT

The MPIAT was established as a result of the September 23, 1999 loss of MCO and the December 3, 1999 loss of MPL and DS2 microprobes. After conducting the MPL and DS2 failure reviews, the MPIAT identified Lessons Learned from various successes and failures within the Mars Program. With these lessons, the MPIAT was tasked with reviewing the revised Mars Surveyor Program to assure Lessons Learned from the previous two failures had been observed and applied at JPL. The MPIAT found the failure to be the result of ineffective implementation of FBC, as opposed to FBC itself, and that cutting testing and validation of system hardware and software was not a legitimate method to decrease cost or reduce timetables. The FBC approach allows for taking acceptable risk in relation to using unproven, high gain technology and not risk associated with the reduction of testing.

In addition, MPIAT recommended that JPL create a Mars Program Office reporting directly to the Laboratory Director and an independent organization at the Directorate level dedicated to implementing major flight projects.

In 2002, an MPL Lesson Learned later described the software failure in three phases. First, in the requirements phase, requirements from NASA Headquarters were not clearly allocated at a system level; the interface with the touchdown sensor did not address transients. Second, the software code and test phase missed a programmer-entered logic error. Further, the simulation did not model touchdown transient signals. Finally, in the system verification phase, an unrelated touchdown sensor wiring issue masked the software issue. The complete end-to-end test was not repeated and the software flaw remained.

Relevance to NASA

Software Assurance

Of particular note is the JPL response to software verification and validation. The official JPL investigation report identified possible physical hazards for all phases of the EDL sequence; however, the one statement that concerned software—common to all EDL phases—read "Flight software fails to execute properly." This broad statement, if applied to hardware, would have solicited more hazard detail at a time when the criticality of software was still being learned.

Other examples attest to a lack of importance placed on critical software assurance (e.g., issues with Space Shuttle Endeavor and the STS-126 crew with a software failure preventing automatic communications hardware reconfiguration; repeated and unexpected Mars Pathfinder system resets; Patriot Missile targeting and tracking failures during deployment in the early 1990s [read about these software issues and more in January 2012 Volume 6 Issue 2 System Failure Case Study *Critical Software: Good Design Built Right* available at nsc.nasa.gov/Articles/SFCS]).

More accurately, software does not fail, it "does not perform as intended." A thorough understanding of software behavior needs to be mapped, including all possibilities for unintended inputs and outputs, because software has a system interface with every aspect of NASA flight hardware.

Verification and Validation

FBC encouraged prudent risk-taking when justified by the return and after extensive review. MPIAT found it to be an effective concept for guiding program implementation. However, risk associated with deviating from sound principals such as thorough verification and validation testing is unacceptable. The MPIAT believed that the FBC concept should be the continued approach utilized in the Mars Program with the stipulation that FBC should not compromise safety.

The Mars '98 project did achieve certain successes with its limited resources, but deviated significantly from a paragon example of analysis and testing. The FBC stratagem involved cheaper, smaller, and more technologically risky spacecraft in higher quantities to hedge losses in case some craft failed. In comparison, NASA's latest victory with Curiosity, the Mars Science Laboratory (MSL) lander, was held back 2 years to complete rigorous software and hardware testing required for the highly advanced mission.

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